

Nishchal Nishant

Machine Learning Engineer — GenAI & MLOps Specialist

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Summary

Machine Learning Engineer with 3+ years of experience building scalable **AI/ML stacks**, **Agentic GenAI systems**, and resilient **MLOps pipelines** at GSK. Expert in Python, SQL, and **Azure Cloud**, with a track record of taking models from experimentation to production. Passionate about designing autonomous multi-agent workflows and optimizing infrastructure costs.

Skills

Languages	Python (Advanced), SQL, Bash
Machine Learning	GenAI, LLMs (GPT-4, Llama), RAG, Multi-Agent Systems, Scikit-learn, TensorFlow, PyTorch
MLOps & Cloud	Azure (AKS, ADF, Functions, AML), Docker, Kubernetes, Airflow, MLFlow, Feast (Feature Store)
Tools	LangChain, Semantic Kernel, Terraform, Git, GitHub Actions, Prometheus, Grafana

Work Experience

Engineer — GSK (EDAP) Jun 2024 – Present

- Market Navigator (GenAI Platform):** Architected an Enterprise GenAI platform to automate marketing product launches. Designed a **multi-agent orchestration layer** (using Semantic Kernel) that autonomously generates, validates, and refines content based on regional data, reducing planning time by 40%.
- Autonomous Infrastructure Agent:** Developed an LLM-based agent to manage user-provisioned Azure resources. The agent validates inputs via REST API and dynamically generates Terraform code to resolve provisioning issues without human intervention.

Senior Associate Engineer — GSK Jul 2022 – Jun 2024

- Synthetic Data Pipeline (Clinical Trials):** Implemented a **CTGAN** pipeline to generate high-fidelity synthetic data from historical clinical trials. Solved critical data scarcity issues for downstream analytics while ensuring strict adherence to patient privacy regulations (GDPR/HIPAA).
- Feature Store Implementation:** Led the POC and implementation of scalable feature stores for the Global Supply Chain. Extensively utilized **Feather** and evaluated Feast/Vertex AI to standardize ML feature engineering across the enterprise.
- MLOps Reliability:** Enhanced the internal data integration tool (**CoDI**) by optimizing Apache Airflow DAGs. Implemented fault-tolerant retry mechanisms that ensured 99.9% job success rates even during Azure pod failures.
- Data Engineering Automation:** Developed automated dashboards and ETL pipelines for the Third Party Security Risk Management (TPSRM) team, significantly reducing manual reporting overhead.

Open Source & Community

DriftDetector — End-to-End MLOps System with Drift-Aware Retraining:

Built a production-ready MLOps pipeline on Azure ML and AKS, enabling real-time anomaly detection, feature store-backed inference, and automated retraining triggered by data drift. Applied MLOps best practices including IaC (Terraform), CI/CD, model versioning, monitoring, and Kubernetes-based scaling.

Machine Learning & AI Knowledge Base

Created and maintained a comprehensive technical documentation repository on GitBook. The project covers deep learning architectures, MLOps best practices, and AI implementation patterns, serving as a reference guide for engineering teams and the open-source community.

[\[View Project\]](#)

Education

2018 – 2022 B.Tech in Information Technology — **BIT Mesra, Ranchi** GPA: 7.9/10.0

Certifications

NLP Specialization (DeepLearning.AI)	Certificate URL
GANs Specialization (DeepLearning.AI)	Certificate URL
Deep Learning Specialization (DeepLearning.AI)	Certificate URL
TensorFlow Developer (DeepLearning.AI)	Certificate URL
Machine Learning on Google Cloud	Certificate URL